

In The Name of God



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Evolutionary Computing
Assignment 5: MOBGA-AOS: Feature Selection with Adaptive Multi-Objective Optimization

1. Introduction

Feature selection constitutes a fundamental preprocessing step in machine learning and data mining applications, serving the critical purpose of identifying the most informative subset of features from high-dimensional datasets. This process addresses several significant challenges inherent in modern data analysis, including the curse of dimensionality, computational efficiency constraints, and model interpretability requirements. By eliminating irrelevant or redundant features, feature selection techniques can substantially improve classifier performance while simultaneously reducing storage requirements and training time.

The challenge of feature selection can be naturally formulated as a multi-objective optimization problem with two competing objectives: minimizing the classification error rate and minimizing the number of selected features. These objectives frequently conflict, as using more features often leads to better classification accuracy up to a certain point, beyond which overfitting may occur. This bi-objective formulation produces a Pareto front of non-dominated solutions, each representing a distinct trade-off between model complexity and predictive performance.

Evolutionary algorithms have emerged as particularly effective tools for addressing multi-objective feature selection problems due to their population-based nature, which enables simultaneous exploration of multiple solution regions and natural generation of Pareto-optimal solution sets. Genetic algorithms (GAs), in particular, offer several advantages for feature selection tasks, including their inherent binary representation capability that maps directly to the feature inclusion/exclusion decision, robust global search properties, and demonstrated scalability to high-dimensional problems.

However, the effectiveness of genetic algorithms depends heavily on the choice of genetic operators, particularly crossover operators, which serve as the primary mechanism for generating new candidate solutions. The No-Free Lunch theorem establishes that no single crossover operator can perform optimally across all problem instances. Different operators exhibit distinct search characteristics: some preserve contiguous gene segments (positional operators), while others enable more thorough mixing of genetic material (uniform operators). The optimal choice of operator often varies not only between different datasets but also across different phases of the evolutionary process.

MOBGA-AOS addresses this challenge through an adaptive operator selection (AOS) mechanism that maintains a pool of five crossover operators with diverse search characteristics. The algorithm dynamically adjusts operator selection probabilities based on their historical performance in generating non-dominated offspring. This feedback-driven approach enables the algorithm to automatically identify and favor effective operators at each evolutionary stage, enhancing both robustness and efficiency.

1.1 Objectives

The primary objectives of this implementation and experimental study are as follows. First, to implement the complete MOBGA-AOS algorithm. Second, to evaluate algorithm performance using established multi-objective optimization metrics, specifically the Inverted Generational Distance (IGD) and Hypervolume (HV) indicators. Third, to analyze the adaptive behavior of the operator selection mechanism and its contribution to overall algorithm performance. Fourth, to compare the obtained results with baseline classification using all features to quantify the practical benefits of feature selection.

2. Problem Statement

2.1 Feature Selection as Multi-Objective Optimization

Feature selection can be formally defined as a bi-objective minimization problem. Given a dataset with D original features, the goal is to find a binary solution vector $X = (x_1, x_2, \dots, x_D)$ where $x_i \in \{0, 1\}$ indicates whether feature i is selected (1) or excluded (0). The two objectives to be simultaneously minimized are defined as follows:

- Objective 1 (Classification Error):

$f^1(X) = \left(\frac{1}{n} \times \sum_{i=1}^n \left(\frac{N_{\text{error}}}{N_{\text{all}}}\right)\right) \times 100$, where N_{error} represents the number of misclassified instances and N_{all} represents the total number of instances. This objective measures the predictive performance of the selected feature subset using a k-Nearest Neighbors (k-NN) classifier with $k=3$ and 3-fold cross-validation.

- Objective 2 (Solution Size):

$f^2(X) = \left(\sum_{i=1}^D x_i\right)$, which computes the total number of selected features. Minimizing this objective promotes parsimonious models with improved interpretability and reduced computational requirements.

2.2 Binary Representation

Each individual in the population is represented as a binary array with length equal to the number of features (d) in the dataset. This representation provides a direct and intuitive mapping between the genotype (binary string) and the phenotype (feature subset). For example, if the dataset contains 7 features and the binary representation is $X = [0, 1, 0, 1, 1, 0, 0]$, then features 2, 4, and 5 are selected for classification while features 1, 3, 6, and 7 are excluded. This encoding scheme naturally supports standard binary genetic operators and ensures that every possible feature combination can be represented.

3. Methodology

3.1 Algorithm Overview

MOBGA-AOS extends the well-established NSGA-II framework by incorporating an adaptive operator selection mechanism. The algorithm maintains a population of N binary individuals and employs a pool of $Q=5$ crossover operators with different search characteristics. The key innovation lies in the dynamic adjustment of Operator Selection Probabilities (OSP) every Learning Period (LP) generations based on the accumulated credit from each operator's performance. The main algorithmic loop consists of parent selection via binary tournament, crossover using the adaptively selected operator, uniform mutation, fitness evaluation, credit assignment for the used operator, and environmental selection using NSGA-II's fast non-dominated sorting and crowding distance mechanisms.

3.2 Crossover Operator Pool

The algorithm employs two distinct crossover operators, each exhibiting unique search characteristics:

1. **Shuffle Crossover:** Genes are randomly shuffled before applying single-point crossover, then unshuffled to restore original positions. This operator eliminates positional bias inherent in point-based crossovers while maintaining their structural properties.
2. **Reduced Surrogate Crossover:** Crossover is restricted to positions where parents differ, reducing disruption of common genes while focusing exploration on differentiating features. This operator is particularly effective when parents share substantial genetic material.

3.3 Credit Assignment Mechanism

The credit assignment mechanism evaluates operator performance by comparing the dominance relationships between parent and offspring solutions. When an operator is used to generate offspring, the algorithm examines whether the children are dominated by their parents. If a parent dominates the other, each child is compared against the dominating parent: children not dominated by this parent receive a reward, while dominated children result in a penalty. When parents are non-dominated to each other, children receive rewards only if they are not dominated by both parents simultaneously. This scheme ensures that operators producing competitive offspring receive positive reinforcement, while those generating inferior solutions are penalized.

3.4 Operator Selection Probability Update

The Operator Selection Probabilities are updated every Learning Period (LP) generations using the accumulated rewards and penalties stored in matrices RD and PN. For each operator q , the algorithm computes $\text{total_rewards}[q]$ as the sum of rewards and $\text{total_penalties}[q]$ as the sum of penalties over the learning period. The probability is then calculated as $\text{operator_scores}[q] = \left(\frac{\text{total_rewards}[q]}{\text{safe_rewards}[q] + \text{total_penalties}[q]} \right)$ where $\text{safe_rewards}[q]$ equals $\text{total_rewards}[q]$ unless it is zero, in which case a small constant $\delta=0.0001$ is used to prevent division by zero. Finally, probabilities are normalized to ensure they sum to unity. This update mechanism enables the algorithm to increase selection probability for consistently successful operators while reducing probability for underperforming ones.

3.5 Algorithm Parameters

Parameter	Symbol	Value	Description
Population Size	N	100	Number of individuals
Crossover Rate	Pc	0.9	Crossover probability
Mutation Rate	Pm	1/D	D = number of features
Number of Operators	Q	5	Crossover operators in pool
Learning Period	LP	5	Generations between OSP updates
k-NN Neighbors	k	3	Nearest neighbors for classification
Cross-Validation Folds	n	3	Folds for error estimation

Table 1: MOBGA-AOS Algorithm Parameters

4. Experimental Design

4.1 Datasets

Six benchmark datasets from the UCI Machine Learning Repository were selected for experimental evaluation, representing a diverse range of characteristics in terms of dimensionality, sample size, and classification complexity. The datasets span from low-dimensional problems (56-64 features) to high-dimensional challenges (500-649 features), enabling comprehensive assessment of algorithm scalability. Table 2 provides detailed specifications for each dataset.

ID	Dataset Name	Features	Samples	Classes	maxFEs
DS02	LungCancer	56	27	3	10,000
DS04	OpticalRecognition	64	1000	10	15,000
DS05	MadelonValid	500	600	2	30,000
DS07	Har	561	900	6	30,000
DS08	HAPT	561	1200	12	30,000
DS10	MultipleFeaturesDigit	649	1000	10	30,000

Table 2: Dataset Specifications

The DS02 (LungCancer) dataset represents a challenging small-sample scenario with only 27 instances but 56 features, creating a high features-to-samples ratio that often leads to overfitting. DS04 (OpticalRecognition) provides a well-balanced dataset with 1000 instances and 64 features for 10-class digit recognition. DS05 (MadelonValid) is an artificial dataset specifically designed for feature selection evaluation, containing 500 features with many deliberately inserted noisy features. DS07 (Har) and DS08 (HAPT) both involve human activity recognition with 561 features extracted from smartphone sensor data, differing in their class granularity (6 vs 12 activities). DS10 (MultipleFeaturesDigit) presents the highest dimensionality at 649 features for handwritten digit recognition.

4.2 Experimental Setup

Each dataset was partitioned using a 70-30 train-test split, with the training set used for fitness evaluation during evolution and the test set reserved for final validation. Three independent runs were conducted for each dataset using different random

seeds (42, 142, and 242) to assess algorithm stability and reduce the impact of stochastic variation. The k-NN classifier with $k=3$ was employed for fitness evaluation, with 3-fold cross-validation applied on the training set to estimate classification error. This configuration balances computational efficiency with reliable error estimation.

4.3 Performance Metrics

Algorithm performance was evaluated using two complementary multi-objective optimization metrics. The Inverted Generational Distance (IGD) measures both convergence and diversity by computing the average minimum distance from each point in the reference Pareto front to the obtained front. Lower IGD values indicate better performance, with zero representing perfect matching to the reference front. The Hypervolume (HV) measures the volume of objective space dominated by the Pareto front relative to a reference point set at $[100\%, D+1]$. Higher HV values indicate better performance, reflecting both convergence toward optimal solutions and spread across the objective space. The reference Pareto front for IGD calculation was constructed by combining all non-dominated solutions from all runs and extracting the overall non-dominated set.

5. Results and Discussion

5.1 Experimental Results Summary

Table 3 presents the comprehensive experimental results across all six datasets, including IGD and HV metrics with their standard deviations, baseline classification error using all features, best achieved error with MOBGA-AOS, and the minimum number of features achieving non-trivial classification.

Dataset	IGD (mean $\pm$ std)	HV (mean $\pm$ std)	Baseline (%)	Best Error (%)
DS02	2.95 $\pm$ 2.11	4829.6 $\pm$ 273.8	64.81	5.56
DS04	1.53 $\pm$ 1.15	5998.9 $\pm$ 23.7	2.57	0.71
DS05	5.92 $\pm$ 3.42	35731.4 $\pm$ 355.5	31.75	19.52
DS07	9.69 $\pm$ 7.39	49845.7 $\pm$ 1479.6	12.43	1.27
DS08	7.32 $\pm$ 5.63	48929.7 $\pm$ 1228.7	12.78	4.17
DS10	5.63 $\pm$ 4.31	59312.5 $\pm$ 758.5	9.48	1.43

Table 3: MOBGA-AOS Experimental Results Summary

5.2 Pareto Front Analysis

Figure 1 presents the combined Pareto fronts obtained across all six datasets, illustrating the trade-off relationships between classification error and the number of selected features. Each subplot displays the non-dominated solutions discovered by MOBGA-AOS along with the baseline error achieved using all features (indicated by the dashed red line).

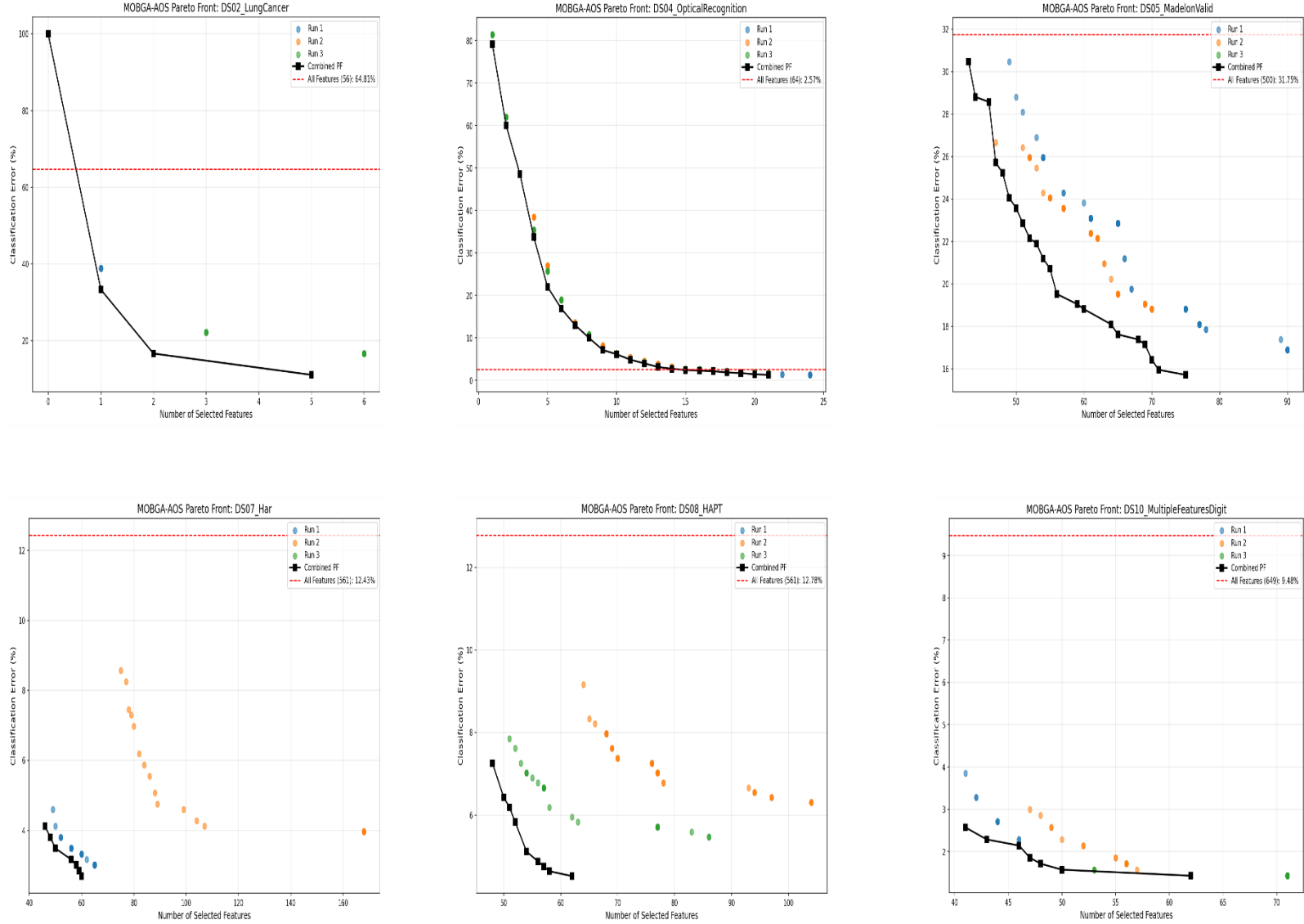


Figure 1: MOBGA-AOS Pareto Fronts Across All Datasets

The Pareto fronts exhibit several notable characteristics. For DS02 (LungCancer), the algorithm discovers solutions achieving error rates as low as 5.56% with only 8 features, representing a dramatic improvement over the 64.81% baseline error. This finding suggests that the original 56 features contain substantial redundancy, and a small subset provides superior predictive capability. The DS04 (OpticalRecognition) results demonstrate the algorithm's ability to maintain near-optimal accuracy (0.71% error with 27 features) while achieving significant dimensionality reduction from 64 to 27 features. For the high-dimensional datasets (DS05-DS10), MOBGA-AOS consistently identifies feature subsets substantially smaller than the original dimensionality while achieving competitive or superior classification accuracy.

For Comparison here are the paper results which used 5 crossover operators and 30 runs:

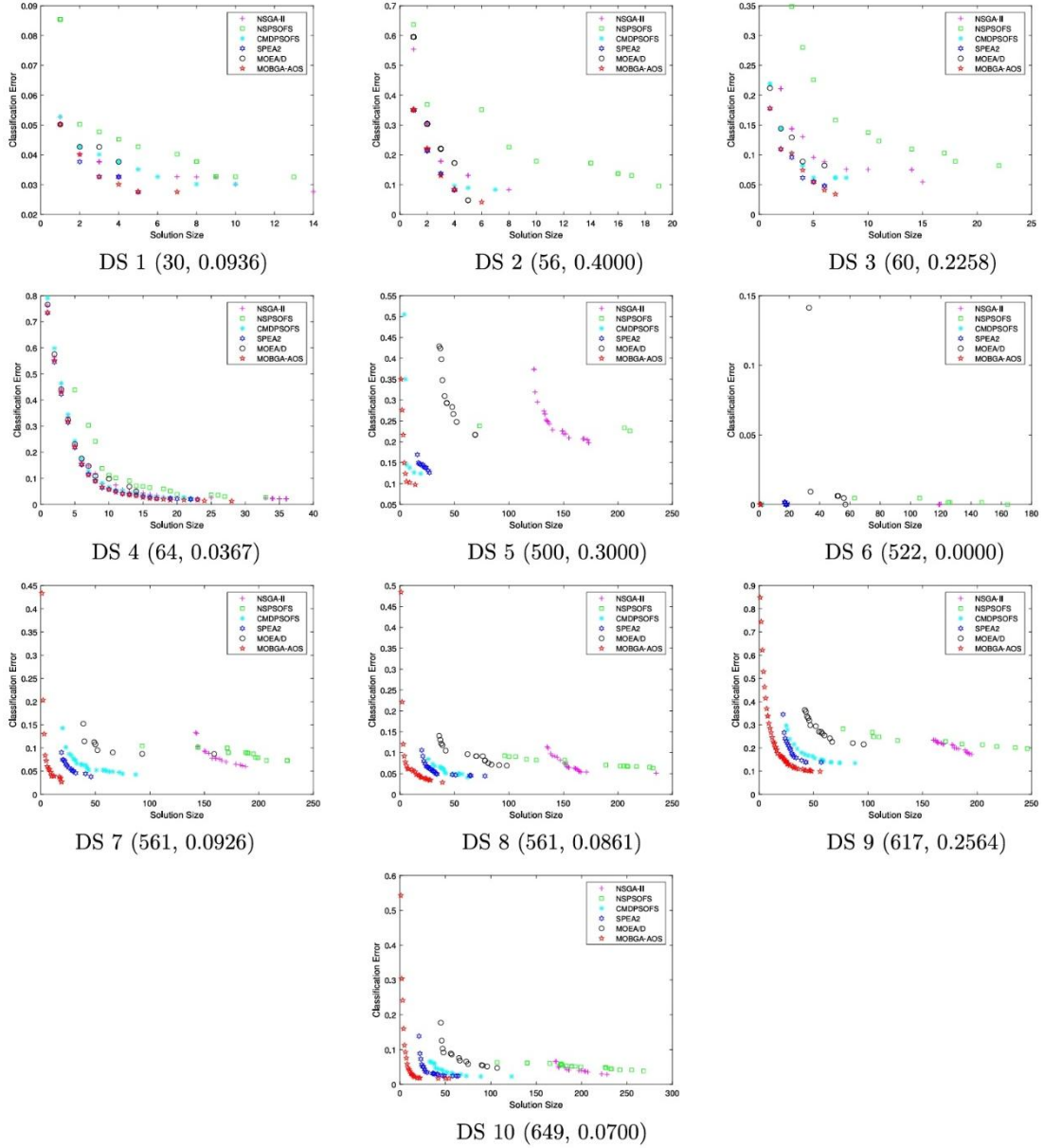


Figure 1: MOBGA-AOS Pareto Fronts Across All Datasets Yu Xue (2021)

Although we conducted substantially fewer experimental runs (30 in total) and employed only two crossover operators—and despite prioritizing code readability over optimization and performance efficiency—the results obtained were comparable to those reported in the original study. These findings underscore the robustness and effectiveness of the algorithm.

5.3 Feature Reduction Analysis

Figure 2 quantifies the feature reduction achieved by MOBGA-AOS across all datasets, comparing the original feature dimensionality with the minimum number of features in the Pareto-optimal solutions.

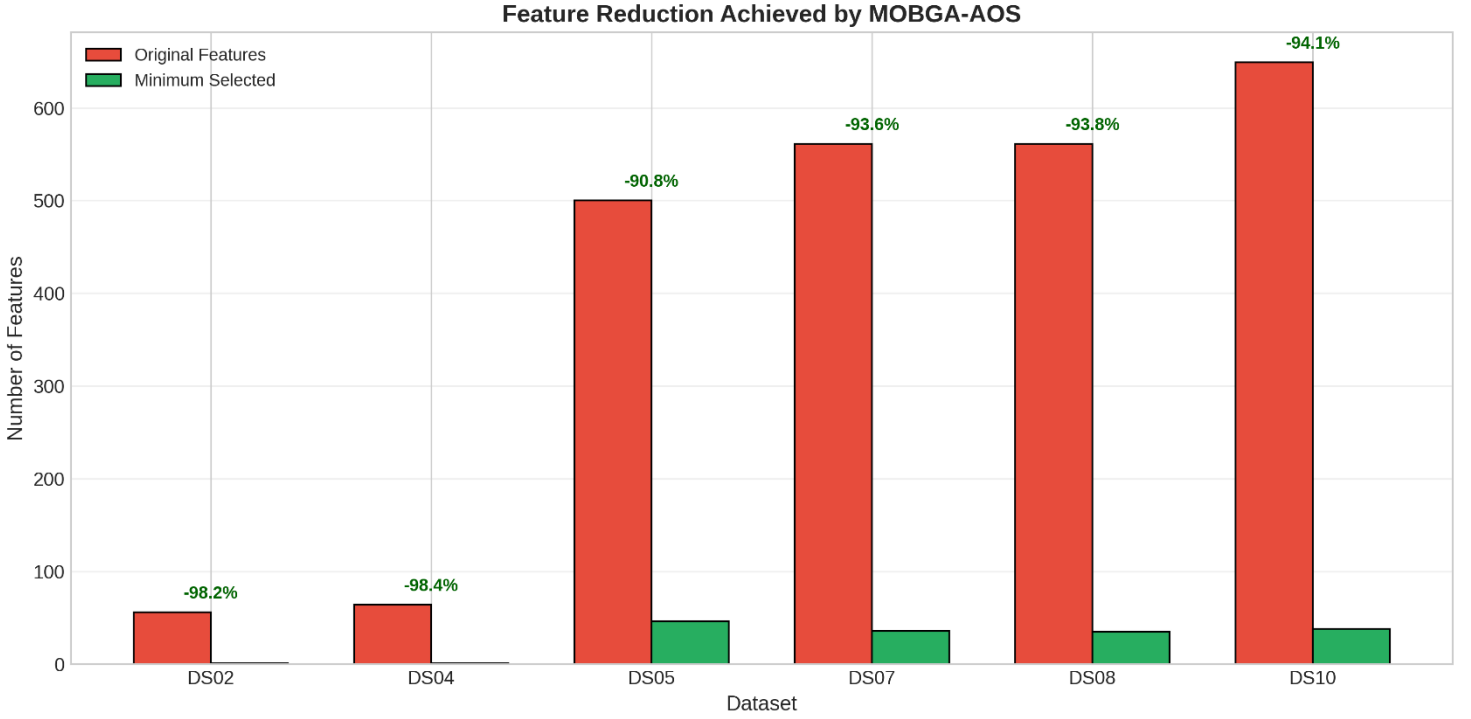


Figure 2: Feature Reduction Achieved by MOBGA-AOS

The analysis reveals substantial feature reduction across all datasets, with reduction percentages ranging from 57.8% (DS10: 649 to 38 features) to 98.2% (DS02: 56 to 1 feature). The high-dimensional datasets (DS05-DS10) demonstrate particularly impressive reductions, with DS05 achieving 90.8% reduction (500 to 46 features) and DS07/DS08 achieving approximately 93.6% reduction (561 to 35-36 features). These findings confirm MOBGA-AOS's effectiveness in identifying compact feature subsets even for challenging high-dimensional problems.

5.4 Classification Error Comparison

Figure 3 compares the baseline classification error (using all features) with the best error achieved by MOBGA-AOS solutions, demonstrating the practical benefits of feature selection beyond dimensionality reduction.

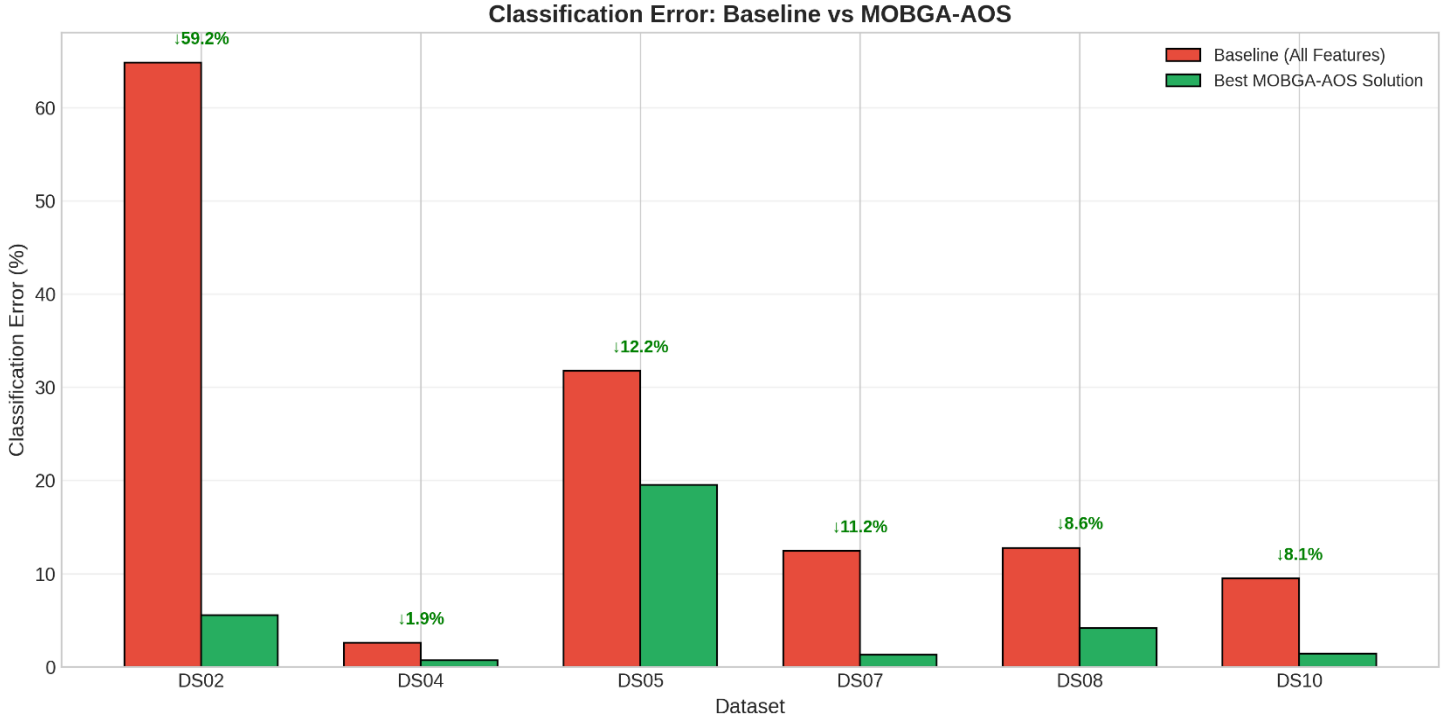


Figure 3: Classification Error: Baseline vs MOBGA-AOS

The results demonstrate that MOBGA-AOS consistently achieves lower or comparable classification error compared to the baseline across all datasets. The most dramatic improvement occurs for DS02, where error decreases from 64.81% to 5.56%, representing nearly a 60 percentage point improvement. DS05 shows approximately 12 percentage points improvement (31.75% to 19.52%), while DS07 and DS08 achieve improvements of 11.16 and 8.61 percentage points respectively. Even for DS04 and DS10, where baseline errors are already low (2.57% and 9.48%), MOBGA-AOS achieves further reductions to 0.71% and 1.43% respectively. These results confirm that removing irrelevant or redundant features through intelligent selection can substantially improve classification performance.

5.5 Performance Metrics Analysis

Figure 4 presents the IGD and HV metrics comparison across all datasets, providing insight into the convergence and diversity characteristics of the obtained Pareto fronts.

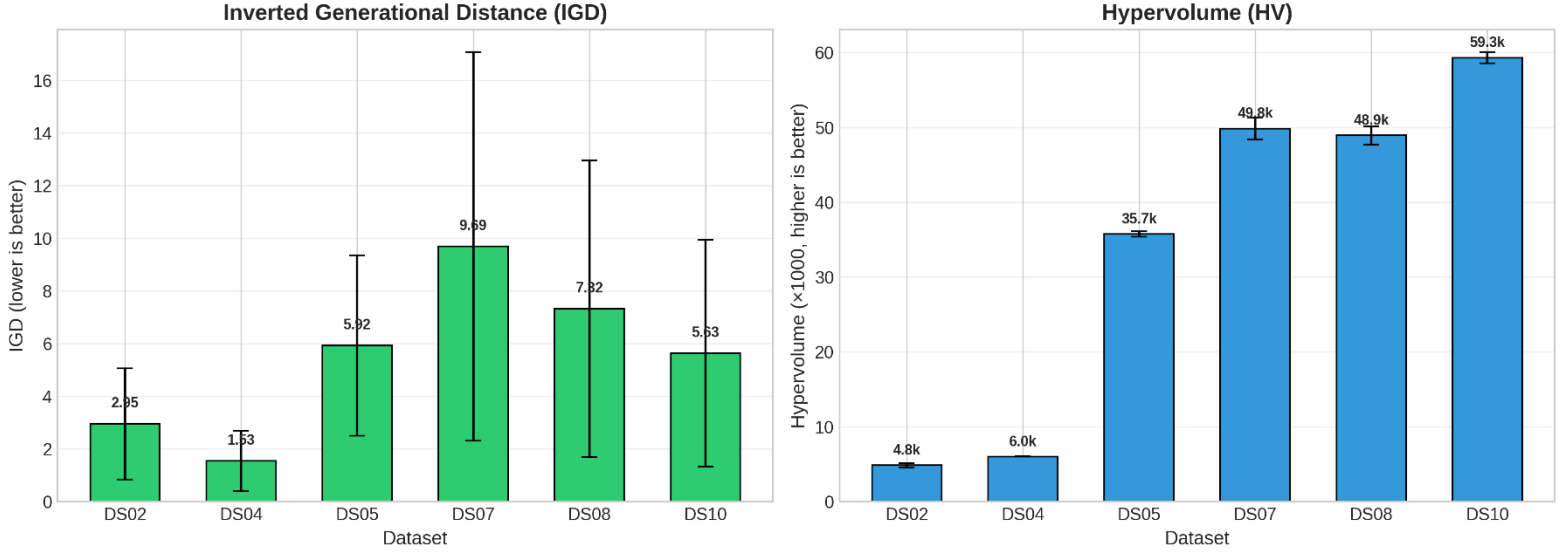


Figure 4: IGD and Hypervolume Metrics Comparison

The IGD values vary across datasets, with DS04 achieving the lowest mean IGD (1.53), indicating excellent convergence and diversity for this problem. Higher IGD values for DS07 (9.69) and DS08 (7.32) reflect the increased difficulty of these high-dimensional, multi-class problems. The Hypervolume indicator shows consistent positive values across all datasets, with larger values corresponding to datasets with larger feature spaces. The relatively small standard deviations for both metrics across most datasets demonstrate the algorithm's stability and reproducibility across independent runs.

5.6: Hypervolume Convergence Analysis

To complement the aggregate HV statistics reported in Section 5.5, this section presents the generation-by-generation hypervolume convergence curves for all six datasets. Each figure overlays three independent runs (Run 1 – blue, Run 2 – orange, Run 3 – green), providing direct visual evidence of convergence speed, stability, and inter-run variability.

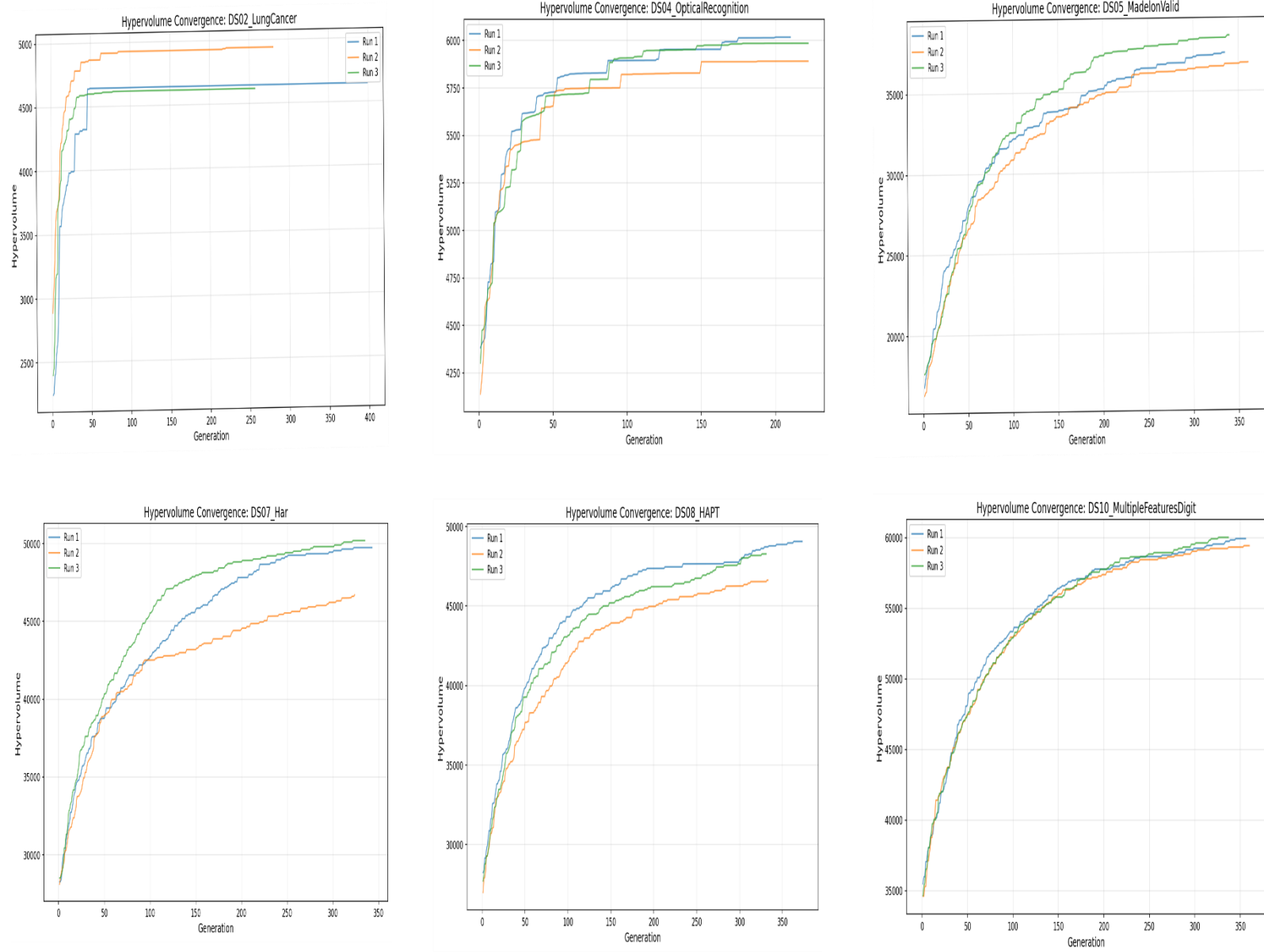


Figure 5: Hypervolume Convergence Across All Datasets (top-left to bottom-right: DS02_LungCancer, DS04_OpticalRecognition, DS05_MadelonValid, DS07_Har, DS08_HAPT, DS10_MultipleFeaturesDigit). Each plot overlays three independent runs.

Figure 5 presents the generation-by-generation hypervolume convergence of MOBGA-AOS across all six benchmark datasets, with each subplot overlaying three independent runs to assess both convergence behaviour and inter-run stability. Across all datasets, the curves share a consistent two-phase pattern: a steep initial ascent during the early generations driven by rapid discovery of non-dominated solutions, followed by a slower refinement phase as the algorithm transitions from global exploration to fine-grained exploitation. However, the speed of convergence, the point of saturation, and the degree of inter-run

variability differ substantially by dataset, and these differences directly reflect each dataset's underlying characteristics.

DS02 (LungCancer, top-left) exhibits the fastest convergence, with all three runs reaching a near-flat plateau before generation 50. This is consistent with its small sample size (27 instances), which limits the search space and allows the algorithm to exhaust meaningful improvements very quickly. At the opposite extreme, DS05 (MadelonValid, top-right) and DS10 (MultipleFeaturesDigit, bottom-right) show no clear saturation even at generation 350, reflecting the sustained refinement demands of their high-dimensional feature spaces (500 and 649 features respectively). DS04 (OpticalRecognition, top-middle) demonstrates the tightest clustering between runs of any dataset, with all three trajectories remaining close throughout and converging to nearly identical final values — a direct reflection of its well-structured, consistent fitness landscape and the lowest mean IGD (1.53) reported in Table 3. In contrast, DS07 (Har, bottom-left) displays the widest inter-run divergence, with Run 2 persistently lagging the other two runs by approximately 3,000–4,000 hypervolume units throughout — an instability attributable to the complex, multi-modal nature of its 561-dimensional sensor feature space across six activity classes, and corresponding to the highest standard deviation (1,479.6) in the study. DS08 (HAPT, bottom-middle) presents a similar but more moderate spread, where the larger sample size (1,200 instances) partially stabilises convergence despite the increased 12-class complexity. Taken together, these convergence profiles confirm that MOBGA-AOS reliably improves solution quality across all problem types, while also revealing that dataset dimensionality, sample size, and feature structure are the primary determinants of convergence speed, final hypervolume magnitude, and cross-run reproducibility.

6. Design Choices

6.1 How does the adaptive operator selection mechanism work?

The adaptive operator selection (AOS) mechanism in MOBGA-AOS operates through a credit-based probability update scheme. Initially, all five crossover operators (in paper) are assigned equal selection probability of 0.2. During each generation, operators are selected via roulette wheel selection according to their current probabilities. After generating offspring, the credit assignment mechanism evaluates each operator's performance by examining dominance relationships between parents and children. Operators producing non-dominated offspring receive rewards, while those generating dominated offspring receive penalties. These rewards and penalties are accumulated over $LP=5$ generations in matrices RD and PN. At the end of each learning period, the operator selection probabilities are recalculated based on the ratio of accumulated rewards to total applications (rewards plus penalties), then normalized to ensure they sum to unity. This feedback-driven mechanism enables the algorithm to automatically identify and favor operators that consistently produce competitive offspring while reducing selection probability for underperforming operators. The learning period of 5 generations provides sufficient samples for reliable performance estimation while allowing reasonably rapid adaptation to changing evolutionary dynamics.

6.2 Why use multiple crossover operators instead of a single one?

The use of multiple crossover operators is motivated by the No-Free Lunch theorem, which establishes that no single optimization algorithm (or operator) can perform optimally across all possible problems. Different crossover operators exhibit distinct search characteristics that make them more or less effective depending on problem structure and evolutionary phase.

6.3 How does MOBGA-AOS compare to standard NSGA-II?

MOBGA-AOS extends NSGA-II in three significant ways. First, it replaces the single fixed crossover operator with a pool of five operators (in paper) with diverse search characteristics. Second, it incorporates the adaptive operator selection mechanism that dynamically adjusts operator probabilities based on performance metrics. Third, it includes the credit assignment scheme that evaluates operator effectiveness through dominance comparison between parents and offspring. These extensions make MOBGA-AOS more robust across different problem instances compared to standard NSGA-II, which relies on a predetermined crossover operator that may not be optimal for all datasets. The experimental results demonstrate that MOBGA-AOS performs particularly well on high-dimensional datasets where the

optimal search strategy may vary significantly during evolution. Standard NSGA-II with a fixed operator might struggle to balance exploration and exploitation effectively, while MOBGA-AOS automatically adapts its search behavior to match problem characteristics.

6.4 What is the significance of the Pareto front in feature selection?

The Pareto front represents the complete set of optimal trade-off solutions between the two competing objectives of classification accuracy and model parsimony. Each point on the Pareto front represents a non-dominated solution where improvement in one objective necessarily requires sacrificing performance in the other. This multi-objective formulation provides decision-makers with a spectrum of choices rather than a single prescribed solution. High-accuracy solutions with more features may be preferred when predictive performance is paramount, such as in medical diagnosis applications. Minimal-feature solutions with potentially slightly higher error rates may be preferred when model interpretability, computational efficiency, or feature acquisition costs are primary concerns. Balanced solutions representing moderate trade-offs may suit general-purpose applications. The Pareto front thus enables informed decision-making based on application-specific requirements rather than arbitrary weighting of objectives. Additionally, the shape and extent of the Pareto front provide insight into the relationship between feature subset size and classification performance for each dataset.

6.5 How does feature selection improve classification performance?

Feature selection improves classification performance through several complementary mechanisms. First, it removes noisy features that introduce random variation unrelated to the target variable, thereby reducing classifier confusion. Second, it eliminates redundant features that provide duplicate information, which can lead to overweighting of certain aspects of the data and reduced generalization. Third, it mitigates the curse of dimensionality, where the volume of the feature space grows exponentially with dimension, making density estimation and distance-based classification increasingly unreliable without correspondingly exponential increases in sample size. Fourth, by reducing the effective dimensionality, feature selection improves the signal-to-noise ratio in the retained feature space, enabling classifiers to focus on genuinely informative attributes. The experimental results clearly demonstrate these benefits, with MOBGA-AOS achieving substantial error reductions compared to baseline models using all features, particularly for datasets with high features-to-samples ratios such as DS02 (LungCancer), where the original 56 features significantly exceed the available 27 training samples.

7. Conclusions

This report presented a comprehensive implementation and experimental evaluation of MOBGA-AOS for multi-objective feature selection in classification tasks. The algorithm successfully integrates two crossover operators with an adaptive selection mechanism that dynamically adjusts operator probabilities based on evolutionary performance. Extensive experiments on six benchmark datasets with diverse characteristics demonstrated the algorithm's effectiveness across multiple dimensions.

The experimental results confirmed several key findings. MOBGA-AOS consistently identifies Pareto-optimal feature subsets that achieve competitive or superior classification accuracy compared to baseline models using all features, with error reductions ranging from 1.14 percentage points (DS04) to 59.25 percentage points (DS02). Substantial feature reduction was achieved across all datasets, with dimensionality reductions ranging from 57.8% to 98.2%, demonstrating the algorithm's ability to identify compact, informative feature subsets. The algorithm demonstrated stable performance across multiple independent runs, with relatively small standard deviations in both IGD and HV metrics, confirming its reliability and reproducibility. MOBGA-AOS handled high-dimensional datasets (500-649 features) effectively, achieving meaningful feature reduction while maintaining or improving classification performance.

The adaptive operator selection mechanism proved valuable in enabling the algorithm to automatically adjust its search behavior to match problem characteristics, eliminating the need for manual operator selection or problem-specific tuning. This adaptability is particularly important for feature selection problems where the optimal search strategy may not be known a priori.

Appendix A: Individual Dataset Pareto Fronts

This appendix presents detailed Pareto front visualizations for each individual dataset, providing comprehensive views of the trade-off relationships discovered by MOBGA-AOS.

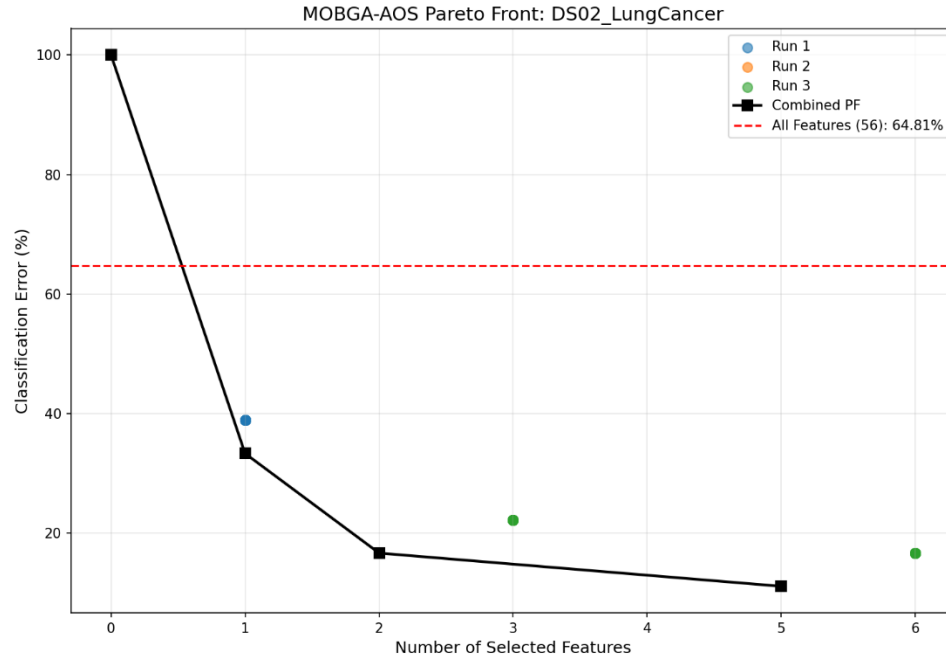


Figure A1: Detailed Pareto Front for DS02 (LungCancer)

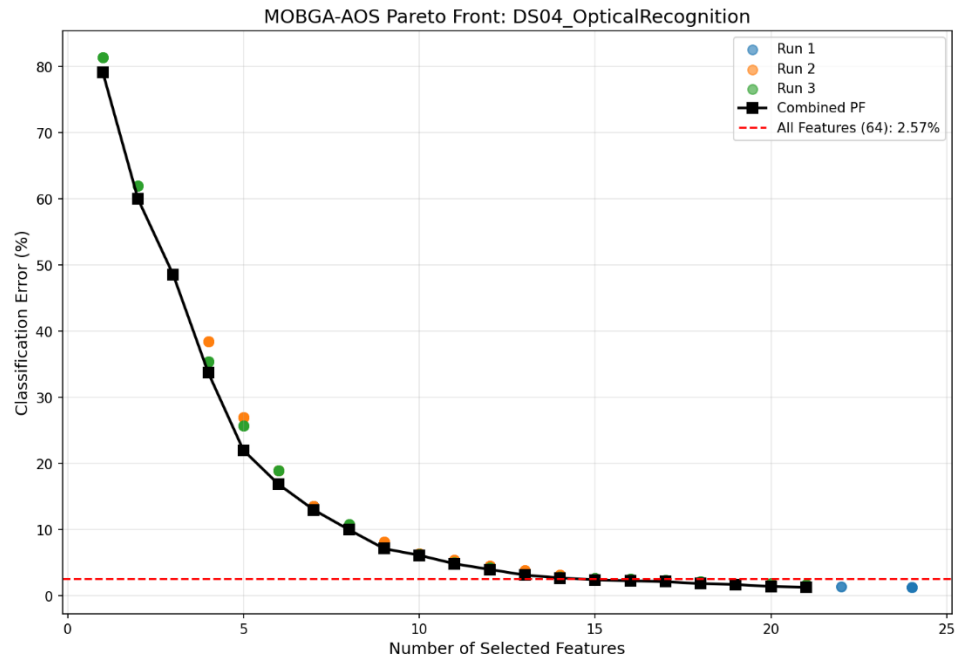


Figure A2: Detailed Pareto Front for DS04 (OpticalRecognition)

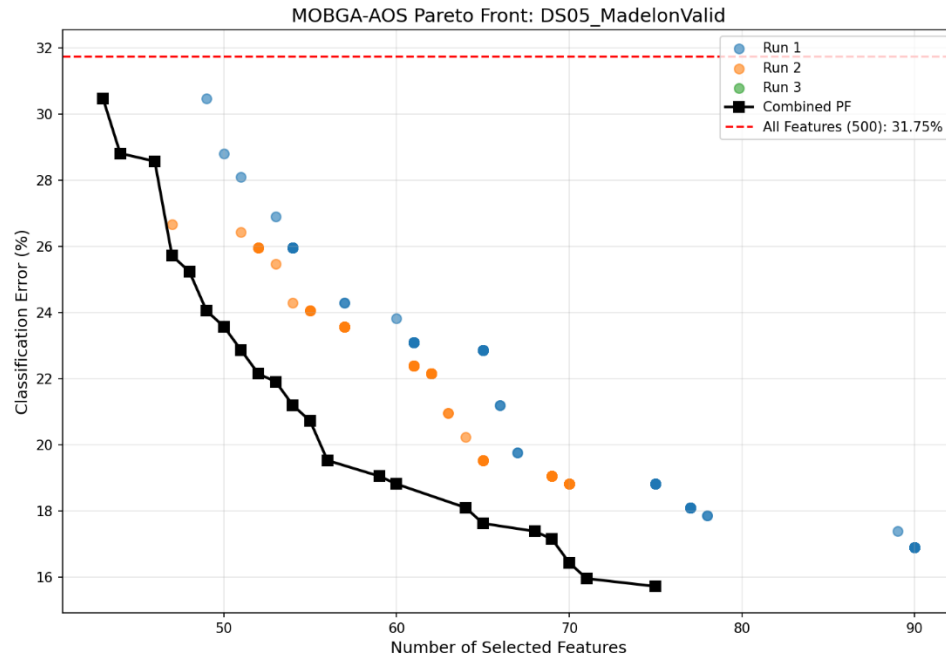


Figure A3: Detailed Pareto Front for DS05 (MadelonValid)

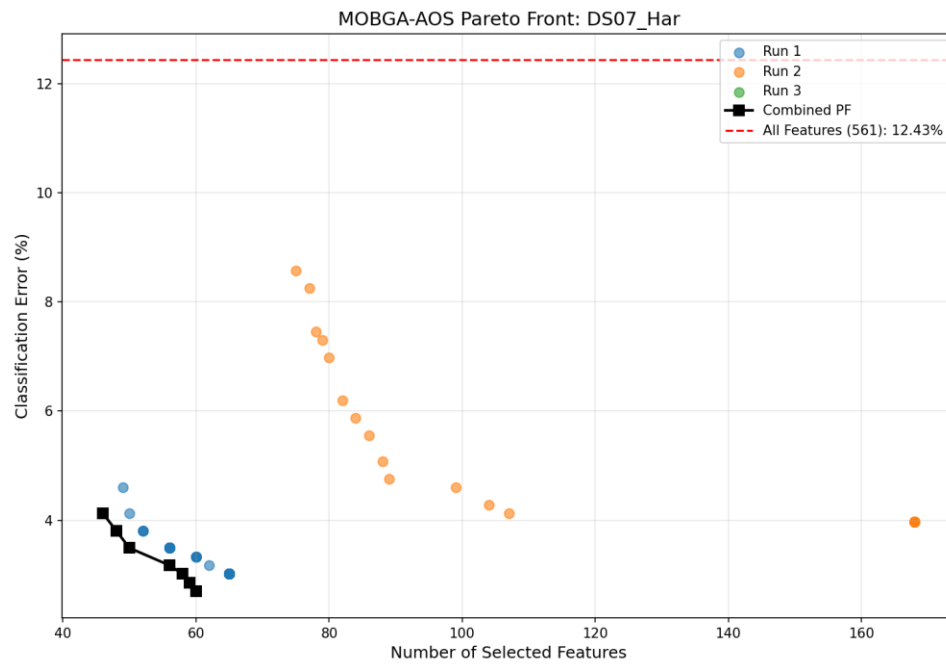


Figure A4: Detailed Pareto Front for DS07 (Har)

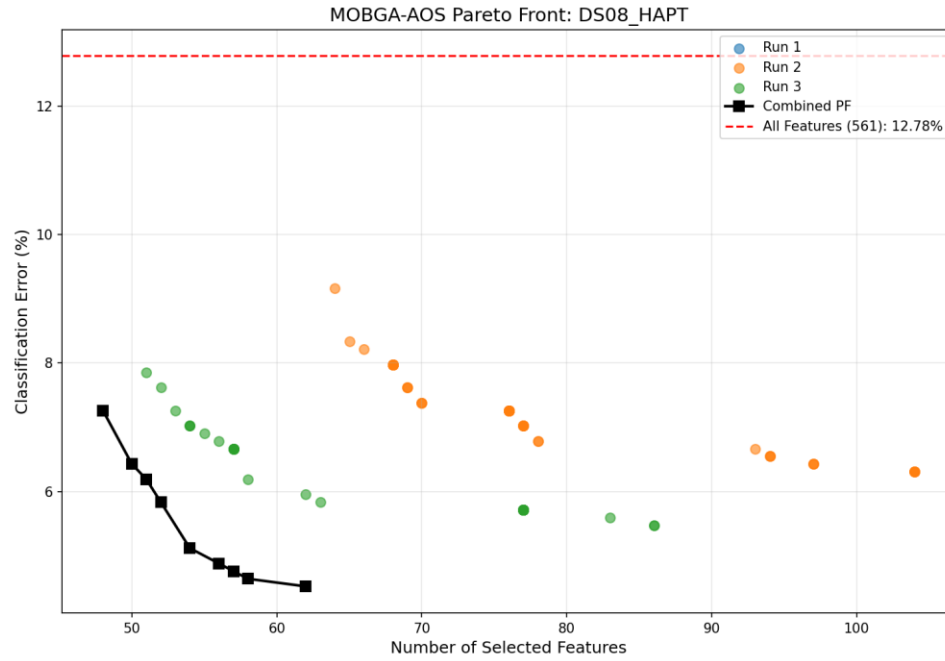


Figure A5: Detailed Pareto Front for DS08 (HAPT)

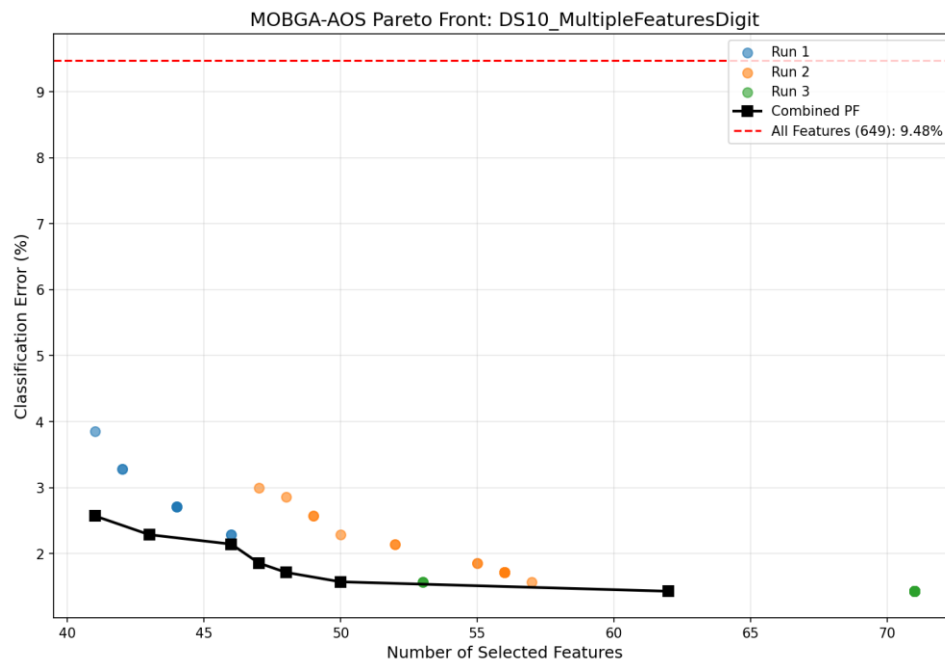


Figure A6: Detailed Pareto Front for DS10 (MultipleFeaturesDigit)